

3.8 Polymerisation

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.8 Polymerisation
Difficulty	Hard

Time allowed: 30

Score: /24

Percentage: /100

Question 1a

Poly(methyl methacrylate) is used as a glass substitute in Perspex.

A section of the polymer chain is shown in Fig. 1.1.

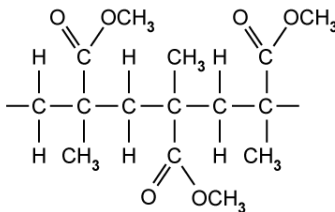


Fig. 1.1

Draw the skeletal formula of the monomer.

[1 mark]

Question 1b

Explain why poly(methylmethacrylate) is fairly unreactive.

[2 marks]

Question 1c

Bioplastics, such as biodegradable plastics offer a sustainable alternative to oil-based products.

Where the use of plant-based polymers is not possible, photodegradable polymers are being developed.

i)

Explain how photodegradable polymers are broken down.

[1]

ii)

Suggest **two** disadvantages of using photodegradable polymers.

[2]

[3 marks]

Question 2a

Explain how p-orbitals are involved in the formation of the C=C bond in chloroalkenes.

You should include a sketch in your answer.

[2 marks]

Question 2b

Compound Y has six possible structural isomers of $C_3H_3Cl_3$.

Two of these isomers are shown in Fig. 2.1.

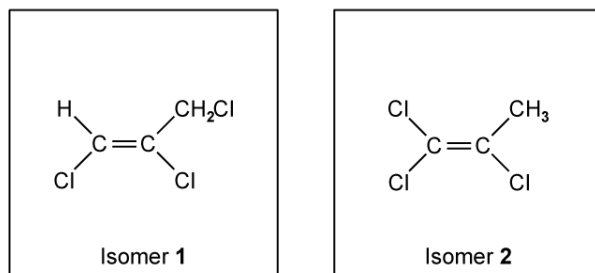


Fig. 2.1

i)

State the IUPAC name of isomer 1.

[1]

ii)

Draw three other structural isomers of $C_3H_3Cl_3$ that are chloroalkenes.

[3]

iii)

Show **two** repeat units for the polymer formed by isomer 2.

[1]

[5 marks]

Question 2c

Polymers can be disposed of by incineration.

i)

Write a balanced symbol equation for the incineration of poly(chloroethene).

Carbon dioxide, water and HCl are formed.

[1]

ii)

Describe the environmental impact of this reaction and explain how it could be reduced.

[3]

[4 marks]

Question 3a

But-2-ene, $\text{CH}_3\text{CH}=\text{CHCH}_3$, is an important compound which is obtained from the cracking of hydrocarbons present in crude oil.

Some reactions of but-2-ene are shown in Fig. 3.1.

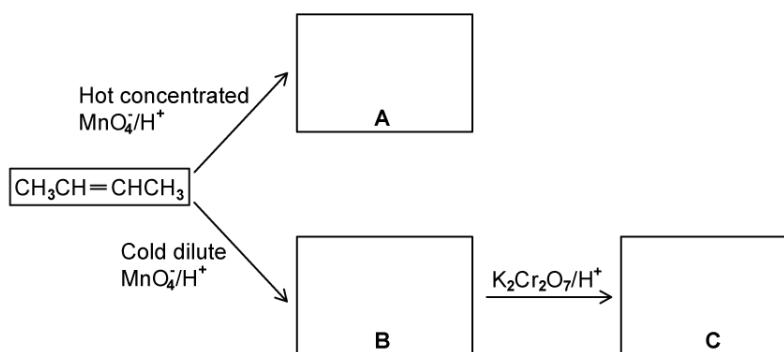


Fig. 3.1

Give the structural formulae of the organic compounds **A** to **C**.

[3 marks]

Question 3b

But-2-ene can be polymerised to give poly(butene). Draw the structural formula of a portion of the polymer chain in poly(butene) showing **two** repeat units.

[1 mark]**Question 3c**

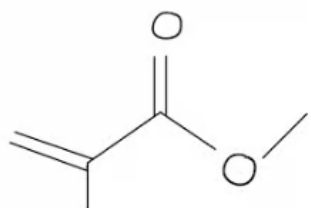
Explain why there is an increasing demand for polymers to be recycled rather than incinerated.

[3 marks]

Model Answers: Hard

1a

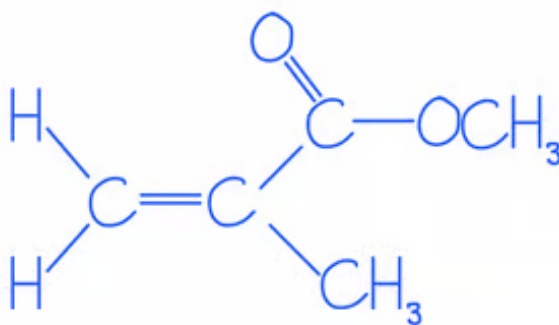
a) The skeletal formula is:



- Correct skeletal formula; [1 mark]

[Total: 1 mark]

- **Remember:** To draw the structure of the monomer, identify the repeat unit in the polymer
- Draw out the displayed formula of the monomer:



- Then convert this into the skeletal formula

1b

b) Poly (methyl methacrylate) is fairly unreactive because:

- It is saturated / contains only single carbon-carbon bonds; [1 mark]

- The hydrocarbon chain is non-polar; [1 mark]

[Total: 2 marks]

- You know from your knowledge of alkanes that alkenes are more reactive due to the presence of the double bond
- Alkanes are fairly unreactive due to being saturated and non-polar, this is the same for polymers

1c

c)

i) Photodegradable polymers are broken down by:

- Absorbing light to weaken the chemical bonds in the polymer; [1 mark]

ii) **Two** disadvantages of using photodegradable polymers are:

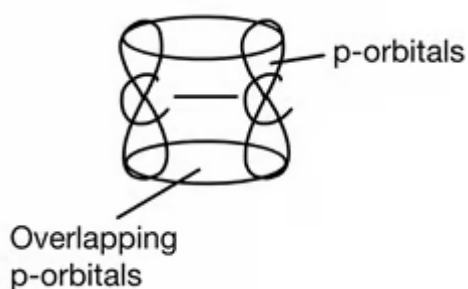
- The polymer may not be exposed to sufficient sunlight; [1 mark]
- They are oil-based; [1 mark]

[Total: 3 marks]

- There are limited questions you can be asked about alternatives to recycling etc. so make sure you're aware of how this process works
- Sometimes, light-absorbing additives are added to polymers to initiate degradation

2a

a) A sketch that shows that p-orbitals are involved in the formation of a double bond is:



- The diagram should have a minimum of 2 carbons, each with a σ bond and p-orbitals; [1 mark]
- There should be an overlap of adjacent p-orbitals (in words or in the diagram); [1 mark]

[Total: 2 marks]

- Questions relating to the involvement of p orbitals in the formation of a double bond will either ask for a sketch or an explanation so make sure you can do both
- **Remember:** Alkanes only have overlapping s orbitals, not p orbitals whereas alkenes have both

2b

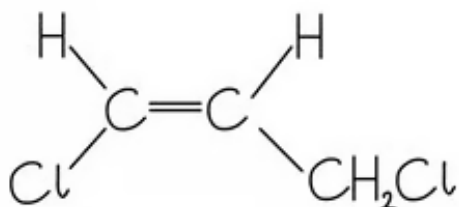
b)

i) The name of isomer 1 is:

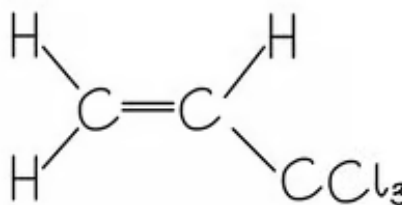
- 1,2,3-trichloropropene; [1 mark]

ii) Three isomers are:

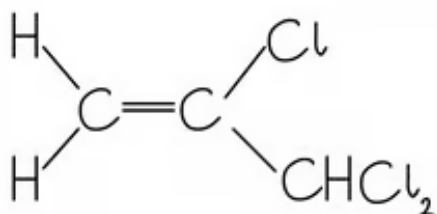
Any **three** from:



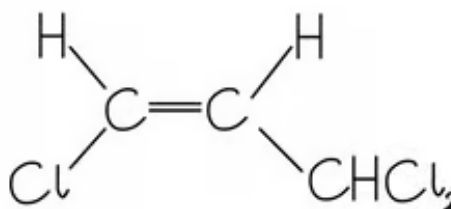
1,1,3-trichloropropene



3,3,3-trichloropropene



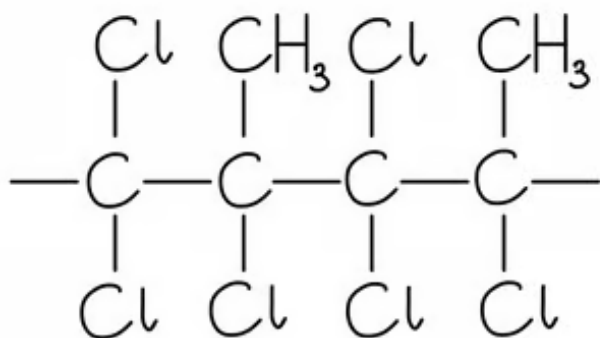
2,3,3-trichloropropene



1,3,3-trichloropropene

- Correct formula for each isomer; [1 mark]

iii) Two repeat units formed by isomer 2 are:



- 4 carbon atoms in the backbone that are connected by single bonds

AND

Correct atoms / groups of atoms bonded to carbon backbone; [1 mark]

[Total: 5 marks]

2c

c)

i) A balanced symbol equation for the incineration of isomer 2 is:

- $-\text{CH}_2\text{CHCl}- + 2 \frac{1}{2}\text{O}_2 \rightarrow 2\text{CO}_2 + \text{H}_2\text{O} + \text{HCl}$; [1 mark]

ii) The environmental impact of incineration includes:

- Carbon dioxide causes global warming; [1 mark]
- Hydrogen chloride is toxic/corrosive; [1 mark]

This could be reduced by:

- Hydrogen chloride can be reacted with a base/alkali/carbonate

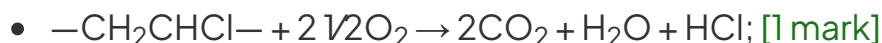
OR

HCl can be neutralised; [1 mark]

[Total: 4 marks]

- Incineration involves burning a substance so you should know poly(chloroethene) reacts with oxygen, and the products are given to you in the question
- Incineration is a problem particularly if the polymer contains chlorine like this one

i) A balanced symbol equation for the incineration of isomer 2 is:



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- Carbon dioxide causes global warming; [1 mark]
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This could be reduced by:

- Hydrogen chloride can be reacted with a base/alkali/carbonate
OR
HCl can be neutralised; [1 mark]

[Total: 4 marks]

- Incineration involves burning a substance so you should know poly(chloroethene) reacts with oxygen, and the products are given to you in the question
- Incineration is a problem particularly if the polymer contains chlorine like this one and PVC
- Make sure you link the pollutant to its effect, you will not be awarded the mark for just stating global warming is a risk if you have not indicated it is caused by carbon dioxide

3a

a) The structural formula of **A**, **B** and **C** are:

- **A** = $\text{CH}_3\text{CO}_2\text{H}$; [1 mark]
- **B** = $\text{CH}_3\text{CH}(\text{OH})\text{CH}(\text{OH})\text{CH}_3$; [1 mark]
- **C** = $\text{CH}_3\text{COCOCH}_3$; [1 mark]

[Total: 3 marks]

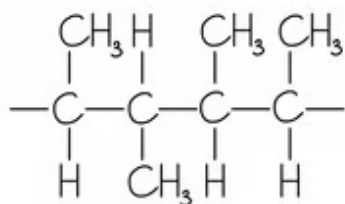
- Alkene will react with acidified potassium manganate(VII)
- The products differ depending on the conditions
- To form compound **A**
 - Hot, concentrated acidified potassium manganate(VII) has been used
 - This breaks the alkene at the C=C bond
 - For but-2-ene the initial product will be ethanal and this will be oxidised further to

ethanoic acid

- To form compound **B**
 - Cold, dilute acidified potassium manganate(VII) has been used
 - This forms a diol when reacted with an alkene
- To form compound **C**
 - The diol has been oxidised to give a ketone
 - In this case, the ketone is butanedione

3b

b) The structural formula containing two repeating units is:



- Two correct repeating units; [1 mark]

[Total: 1 mark]

- You can draw the CH₃ groups in any orientation for these repeating units

3c

c) There is an increasing demand for polymers to be recycled rather than incinerated because:

- Less pollutants / greenhouse gases / CO₂ are released; [1 mark]
- Less energy is required to recycle polymers; [1 mark]
- Oil supplies / fossil fuels / raw materials are conserved; [1 mark]

[Total: 3 marks]

- Polymer disposal is a continuing worldwide problem
- The three main methods of waste disposal are burning, recycling and burying in landfill
- You need to know the advantages and disadvantages of each method as well as be able to compare methods